ESR-DMNet: Enhanced Super-Resolution-Based Dual-Path Metric Change Detection Network for Remote Sensing Images With Different Resolutions

Xi Li, Li Yan, *Member, IEEE*, Yi Zhang, and Huaien Zeng

***Abstract*— Remote sensing change detection (RSCD) has always been one of the research hot issues in remote sensing. Current research focuses on studying deep learning (DL) change detection (CD) methods for remote sensing images with the same resolution. With the prevalence of multiresolution remote sensing images, how to effectively utilize remote sensing images with different resolutions for CD is a key issue. To solve this problem, this article proposes an enhanced super-resolution-based dual-path metric CD network (ESR-DMNet) to realize high-accuracy and high-efficiency end-to-end CD of remote sensing images with different resolutions. ESR-DMNet provides a new enhanced super-resolution (SR) module for the CD of remote sensing images with different resolutions, which can perceptively reconstruct low-resolution (LR) images into more realistic high-resolution (HR) images. ESR-DMNet proposes an effective and efficient dual-path metric CD network, which processes shallow spatial details information and deep semantic information separately to achieve high-accuracy and high-efficiency CD. Compared with nine state-of-the-art methods, our method shows good performance at three resolutions on three datasets, Sun Yat-Sen University (SYSU), season-varying CD dataset (CDD), and cropland CD dataset (CLCD), confirming its potential for CD tasks in remote sensing images with different resolutions.**

***Index Terms*— Change detection (CD), deep learning (DL), different resolution image, metric learning, remote sensing, super-resolution (SR).**

I. Introduction

CHANGE detection (CD) is the process of identifying the state differences by observing the same object or phenomenon at different times. Remote sensing CD (RSCD) has always been one of the research hot issues in RS. Over the past few decades, CD using bitemporal images has been extensively researched. Before the recent explosive development of deep learning (DL), most methods solve the CD problem by manually designing complex feature extractors, which require a large amount of expert knowledge, low detection accuracy, and high computing cost.

In recent years, the number of DL methods applied to RS tasks [1], such as land use and land cover (LULC) classification [2], object detection [3], [4], semantic segmentation [5], and so on, has grown exponentially due to the widespread attention in the rapid development of artificial intelligence (AI). Unlike these RS tasks, CD deals with bitemporal or multitemporal RS images simultaneously. At the same time, the RSCD task [6] has also begun to widely use neural networks with full convolutional neural networks (FCNs). RSCD methods based on FCNs are roughly divided into three architectures [7]: the first is the classification learning RSCD method based on semantic segmentation network, that is, early fusion method. The semantic segmentation network can be directly used for RSCD by merging the bitemporal images into one input using concatenation or differential strategies [8]. The second is the classification learning RSCD method based on the Siamese network, that is, the later fusion method. Use Siamese feature extraction network to extract the bitemporal image features independently, which are merged and fed into another differential discrimination network [9], [10]. The third is metric learning RSCD method based on the Siamese network. Input the bitemporal image into the Siamese network, integrate feature extraction and discriminant implicit metric module [11], increase the gap between unchanged pairs and changes pairs, and measure the distance between features end-to-end to detect changes [12], [13].

Relatively, the metric learning RSCD method based on the Siamese network is less studied and still has some problems. Both low-level details and high-level semantics are critical to RSCD. Low-level visual information is critical to accurately predict the boundaries or details, and high-level semantic features are helpful for category recognition of image regions. One method uses dilated convolution or multiscale blocks [14], [15] to obtain larger receptive fields, but the computational cost is high. Another method exploits middle layer features [16] to generate high-resolution (HR) predictions and encode high-level semantic information. Some methods introduce attention mechanisms, such as self-attention, spatial attention, and channel attention [13], [17]. Although a certain degree of spatial information can be recovered,

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most of the spatial information lost in the deep semantic features cannot be easily recovered by involving the shallow layer [18], [19].

The current stage concentrates on studying DL CD methods under the same resolution RS images. With the prevalence of multiresolution RS images, how to effectively utilize different resolution RS images to detect changes is a key issue. The emergence of super-resolution (SR) [20] based on DL has brought new solutions to different resolution RS image CD. The single image SR task introduces adversarial learning strategies to improve the reconstruction quality in perception and recover more realistic and natural texture details [21], [22]. Single image SR is widely used in RS tasks, such as scene classification and object detection [23], but it is rarely involved in CD tasks.

In response to the above problems, an enhanced SR-based dual-path metric CD network (ESR-DMNet) for different resolution RS images is proposed by combining SR methods and CD methods. ESR-DMNet can perceptually reconstruct low-resolution (LR) images into more realistic HR images and realize the simultaneous acquisition of spatial detail information and semantic information, thus realizing end-to-end high-accuracy and high-efficiency CD for different-resolution RS images. Our main contributions are summarized as follows.

1) We propose an ESR-DMNet to high-accuracy and high-efficiency detect change end-to-end for RS images with different resolutions.
2) We provide a novel enhanced SR (ESR) module for the CD of RS images with different resolutions, which perceptually reconstructs the LR images into more realistic HR images and solves the spatial resolution mismatch problem in CD of different resolution RS images.
3) We propose an effective and efficient dual-path CD architecture, called dual-path metric CD network (DMNet), which processes shallow spatial detail information and deep semantic information separately.

The rest of this article is organized as follows. The related works are reviewed in Section II. Section III describes the proposed network. Section IV introduces the datasets used and implementation details. Section V analyzes the experimental results. Section VI gives the final conclusions.

Algorithm 1 Flowchart of ESR-DMNet

Input:
HR image at time T1 T_1^{hr}, LR image at time T2 T_2^{lr}, HR image at time T2 T_2^{hr}, and ground truth value L.

Function ESR-DMNet(T_1^{hr}, T_2^{lr}, T_2^{hr}, L):
ESR module:
Generate SR image at time T2, T_2^f = ESR module (T_2^{lr}).
DMNet:
$a1$ = Spatial detail branch (T_1^{hr}),
$a11$ = Upsample($a1$),
$a2$ = Semantic branch (T_1^{hr}),
$b1$ = Spatial detail branch (T_2^f),
$b11$ = Upsample($b1$),
$b2$ = Semantic branch (T_2^f),
a = Dual-path FFM ($a11$, $a2$),
b = Dual-path FFM ($b11$, $b2$),
Calculated distance map, $D1$ = Metric module (a, b),
$D2$ = Upsample($D1$),
Get the final change map, $Pred_L = (D2 > 1)$.
Calculate the loss:
Calculating the loss of DMNet, $Loss_{BCL}$,
Calculate the relative discriminator loss, $Loss_D^{re}$,
Calculate the generator losses, $Loss_G$.

Output:
Change map, $Pred_L$.

II. Related Work

A. CD of RS Images With Different Resolutions

Most CD papers focus on the comparison between given image pairs with the same spatial resolution [24], but there are few studies on CD methods of different resolution RS images. In the task of CD in RS images with different resolutions, although the ground objects are semantically consistent [25], they are often incommensurability superficially due to spectral differences [26] and resolution differences [27] arising from different meteorological conditions, different seasons, and so on, during the imaging process. In different resolution RS image CD tasks, the spatial resolution of bitemporal RS images does not match, and the existing CD methods for RS images with the same resolution cannot be directly applied [28], [29], [30]. Some methods use simple prediction-based SR methods (e.g., linear, bilinear, and bicubic interpolation (BI) [31]) to solve the problem of spatial resolution mismatch in bitemporal RS images, choosing to downsample an HR image to match an LR image [32] or to upsample an LR image to match an HR image [24]. The former will lose the detailed spatial information of the HR image [29], [33], and the latter often generates overly smooth images with blurred edges and jagged artefacts [21], [34]. At this point, DL-based SR is introduced to reconstruct the LR image, which has a strong ability to recover the semantic information of the image. Liu et al. [35] integrate SR and Siamese structure to propose an end-to-end Siamese generative adversarial network (GAN) for CD at different scales. Liu et al. [29] propose a CD network with stacked attention modules based on SR to overcome resolution difference between bitemporal images. Liu et al. [28] use multimodal transformers to learn feature alignment to achieve CD with different resolutions. One approach uses SR and unsupervised learning to mitigate spatial and spectral differences in multisensor CD [36]. Another approach proposes a multifeature fusion SR building CD framework based on DL for multisensor images [33], but single-class CD and unsupervised CD are out of our consideration.

Different from the above methods, we propose a dual-path architecture, which integrates low-level details information and high-level semantic information and sets separate spatial information branches with larger space size to achieve the optimal tradeoff between detection accuracy and inference speed. Meanwhile, our proposed method can perceptually

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reconstruct LR images into more realistic HR images and achieve high-accuracy and high-efficiency CD.

B. Super-Resolution

Convolutional neural networks (CNNs) methods have brought prosperity to SR since SRCNN was proposed by Dong et al. [37]. An endless number of network designs and training strategies continue to improve SR performance, especially peak signal-to-noise ratio (PSNR). However, PSNR-oriented methods output results that are transitionally smooth and lack sufficient high-frequency detail because the PSNR disagrees with subjective human assessments [21]. Several researchers have proposed perception-driven approaches to improve the visual quality of SR results. Ledig et al. [21] propose a GAN for image SR (SRGAN), in which a deep residual network and perceptual loss are used to generate photo-level realistic images. However, there is still a pseudoshadow on the image. To further improve image quality visually, Wang et al. [22] propose an enhanced SRGAN (ESRGAN) by including a residual dense block without batch normalization (BN) in the generator and training the network using a relative GAN.

Based on this, a new ESR is introduced for the CD of different resolution RS images, which reconstructs LR images into SR images with better perceptual quality, further improves the accuracy of CD, and solves the dilemma of CD of different resolution RS images.

III. Methodology

In this section, we will introduce the proposed network architecture in detail, including the overall structure of the network, the ESR module, the DMNet, and the loss function.

A. Overview

ESR-DMNet consists of ESR and DMNet (see Fig. 1). The ESR is composed of generator and relative discriminator, which is used to reconstruct LR image into SR image and can recover more semantic information and more realistic details. DMNet includes semantic branch, spatial detail branch, dual-path feature fusion module (FFM), and metric module, which can aggregate both detail information and semantic information. The semantic branch architecture belongs to the family of the encoder–decoder methods, see the purple cuboid in Fig. 1. Four lightweight multipath refinement network (LRNet) modules are used to fuse the rough high-level semantic features with the more fine-grained low-level features to generate HR semantic feature maps. Moreover, three convolution layers are used as spatial detail branch to preserve large spatial size and encode rich spatial information, as shown in the blue cuboid in Fig. 1. We use the dual-path FFM to fuse the output features of these two paths. The output bitemporal feature is input to the metric module, and the change is detected by directly measuring the distance map. ESR-DMNet can achieve both high accuracy and high efficiency.

B. ESR Module

The ESR module is used to reconstruct the LR image at time T_2 into an SR image with better perceptual quality. The ESR is inspired by SRGAN [21] and ESRGAN [22], including generator and relative discriminator.

A new generator [22] is introduced for CD in RS images with different resolutions, and residual-in-residual dense block (RRDB) combining multilevel residual network and dense connectivity is used to improve the reconstruction quality of the SR images. The generator [see Fig. 2(a)] mainly consists of four 3×3 convolutions, 16 RRDBs, and one upsampling. RRDB [see Fig. 2(b) and (c)] employs residual scaling to prevent instability, as well as smaller initialization for easier training.

A new relative discriminator [21], [22], [38] is introduced in different resolution RS images CD. It learns to judge "whether one image is more real than another." This improvement helps the generator recover more realistic texture details. The relative discriminator predicts relative authenticity rather than absolute authenticity

$$D_{\mathrm{re}}\left(T_2^{\mathrm{hr}}, T_2^{f}\right) = \sigma\left(D\left(T_2^{\mathrm{hr}}\right) - E\left[D\left(T_2^{f}\right)\right]\right) \to 1, \quad \text{more realistic than fake data} \tag{1}$$

$$D_{\mathrm{re}}\left(T_2^{f}, T_2^{\mathrm{hr}}\right) = \sigma\left(D\left(T_2^{f}\right) - E\left[D\left(T_2^{\mathrm{hr}}\right)\right]\right) \to 0, \quad \text{less realistic than real data} \tag{2}$$

where D_{re} is a relative discriminator, σ is the sigmoid function, and D is the discriminator [21] [see Fig. 2(d)]. The discriminator contains eight 3×3 convolution, using the BN and the LeakyReLU (LReLU), an AdaptiveAvgPool, two 1×1 convolution, and a final sigmoid activation function to obtain the probability of sample classification. T_2^{hr} is the HR image at time T_2. $T_2^{f} = G(T_2^{\mathrm{lr}})$ is the SR image at time T_2. G is the generator. T_2^{lr} is the LR image at time T_2, and E is for averaging.

C. DMNet

DMNet is proposed to process shallow spatial details separately from deep semantic information and achieve high accuracy and high efficiency. DMNet includes spatial detail branch, semantic branch, dual-path FFM, and metric module, as shown in Figs. 1 and 3.

1) Spatial Detail Branch: Spatial detail branch [19] is responsible for providing low-level spatial details. As shown in the blue dotted box in Fig. 3, the spatial detail branch consists of three layers, each of which contains a convolution, BN, and ReLU. The final output of the three-layer feature map is 1/2, 1/4, and 1/8 of the original image. The three-layer feature map preserves a large spatial size and can provide rich spatial information. This path is set to shallow layer, not computationally intensive, and high efficiency.

2) Semantic Branch: Parallel with spatial detail branch, semantic branch aims to capture high-level semantic information, which has a deeper structure. Semantic branch can capture rich contextual information and large receptive field, inspired by Lin et al. [39] and Nekrasov et al. [40]. As shown

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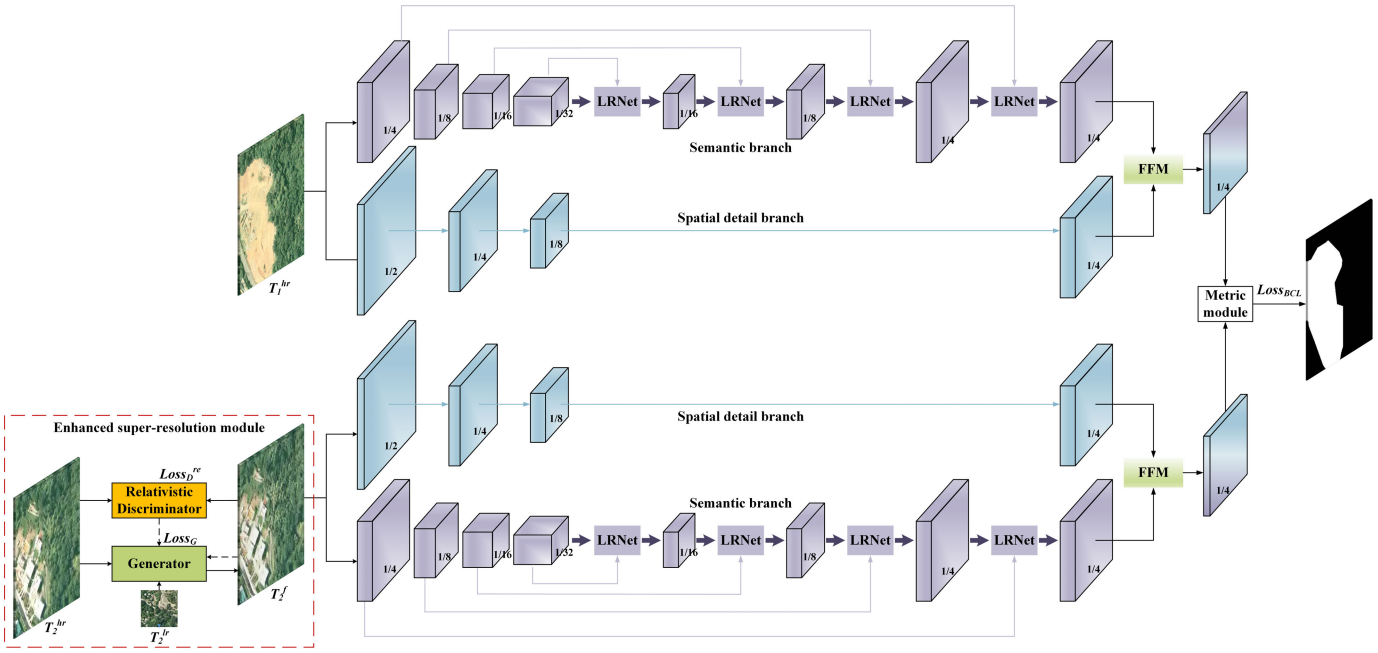


Fig. 1. Illustrative workflow of the proposed ESR-DMNet for CD of RS images with different resolutions. The red dotted box is the ESR module and the remaining is the DMNet. The numbers next to the cuboid display the ratio of the feature map to the size of the input image.

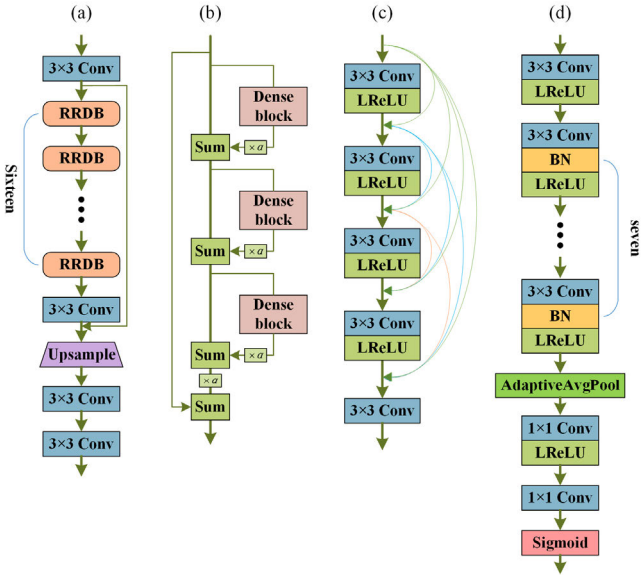


Fig. 2. ESR module. (a) Generator. (b) RRDB, α is residual scale parameter. (c) Dense block. (d) Discriminator.

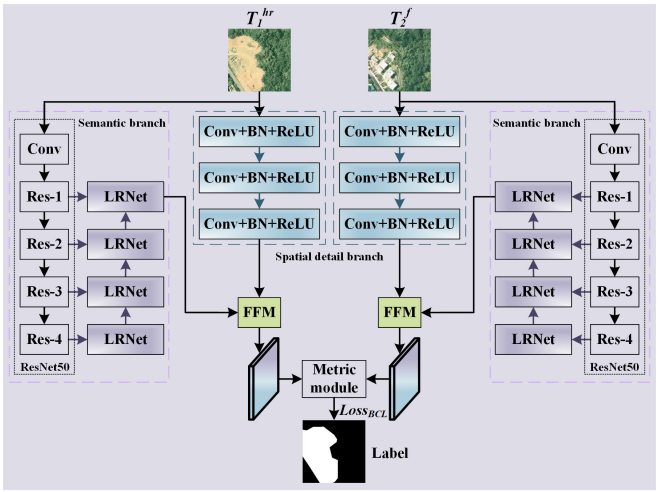


Fig. 3. DMNet.

in the purple dotted box in Fig. 3, a total of four features from Res-1 to Res-4 are extracted using the pretrained ResNet50, respectively, connected to four LRNet, and the captured deep high-level semantic features are directly refined using early convolved fine-grained features. Integrate information from different resolutions and potential remote connections to generate HR semantic feature maps. Although the semantic branch is designed as a deep layer, the gradient can be propagated directly through short-range and long-range residual connection with no effort to generate effective and efficient HR prediction.

As shown in Fig. 4(a), LRNet mainly consists of multiresolution fusion (i.e., Sum) and chained residual pooling (CRP) [39], [40]. The LRNet uses convolution to adjust the dimension of the feature map, multiresolution fusion to fuse two-level feature maps, CRP to capture rich background context information, and upsampling to adjust the feature map size. As shown in Fig. 4(b), CRP consists of four pooling blocks, each of which consisting of a maximum pooling layer and a convolution layer. The output feature maps of all pooling blocks are fused together with the input feature maps by the summation of residual connections. CRP allows access to elements from large regions without using large pooling windows. As shown in Fig. 3, the LRNet connected to Res-4 has only one feature map input, eliminating the sum operations.

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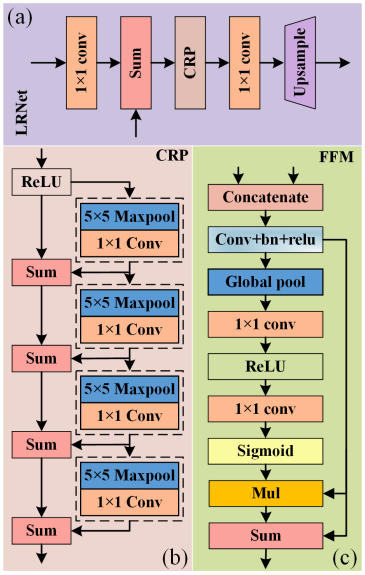


Fig. 4. Architecture of LRNet. (a) LRNet. (b) CRP. (c) Dual-path FFM.

In terms of accuracy, the spatial detail branch provides powerful spatial information, while the semantic branch captures rich background context information and large receptive field. In terms of efficiency, although the spatial size of the spatial detail branch is large, it has only three convolution layers and is not computationally intensive. Semantic branch is implemented using a lightweight model. The spatial detail branch and semantic branch are calculated simultaneously, which further improves the efficiency.

3) Dual-Path FFM: The spatial detail branch is set to shallow layer to provide low-level spatial details. The semantic branch has a deeper structure to capture high-level semantic information. There are semantic and resolution gaps between dual-path features. The dual-path FFM [19] is introduced to select and combine features output from spatial detail branch and semantic branch, as shown in Fig. 4(c).

4) Metric Module: The metric module in Figs. 1 and 3 uses Euclidean distance to directly measure the bitemporal feature maps to obtain the distance map. The batch-balanced contrastive loss (BCL) [12] is used to expand the distance between the changed features and reduce the distance between the unchanged features. The distance map is segmented using a simple threshold to obtain the CD result.

D. Loss Function

We introduce a new perception loss function and a relative discriminator loss [21], [22] in different resolution RS images CD, which is helpful for SR images to learn more sharp edges and more detailed textures. Three loss functions, Loss_{BCL} [12], $\text{Loss}_D^{\text{re}}$, and Loss_G [21], [22], are used to optimize the DMNet, relative discriminator, and generator in the ESR module.

1) DMNet Loss: BCL is introduced to calculate the loss of DMNet, which can enlarge the distance difference between unchanged pairs and changed pairs in the metric module

$$\text{Loss}_{\text{BCL}}(y,d)=\frac{1}{2}\left[\sum_{b,i,j}\frac{1}{n_{b,u}}(1-y_{b,i,j})d_{b,i,j}^2+\sum_{b,i,j}\frac{1}{n_{b,c}}y_{b,i,j}\max\left(m-d_{b,i,j},0\right)^2\right]. \tag{3}$$

The binary label $y_{b,i,j}\in\{0,1\}$, $d_{b,i,j}$ is the distance between the bitemporal feature maps. b, i, and j indicate the indexes of batch, height, and width, respectively. $n_{b,u}$ and $n_{b,c}$ are the number of unchanged and changed pixel pairs, respectively. Values greater than m do not participate in the calculation of Loss_{BCL}.

2) Relative Discriminator Loss: A relative discriminator loss is introduced into the CD task of RS images with different resolutions, which helps to learn sharper edges and more detailed textures

$$\text{Loss}_D^{\text{re}}=-E\left[\log\left(D_{\text{re}}\left(T_2^{\text{hr}},T_2^f\right)\right)\right]-E\left[\log\left(1-D_{\text{re}}\left(T_2^f,T_2^{\text{hr}}\right)\right)\right] \tag{4}$$

$$\text{Loss}_D=\lambda\text{Loss}_D^{\text{re}}. \tag{5}$$

3) Generator Loss: The generator loss includes three parts, namely, perception loss, adversarial loss of generator, and content loss. A new perceptual loss is introduced into the CD task of RS images with different resolutions, which provides clearer edge and better visual perception

$$\text{Loss}_G=\text{Loss}_{\text{percep}}+\beta\text{Loss}_G^{\text{re}}+\delta\text{Loss}_{\text{cont}} \tag{6}$$

$$\text{Loss}_{\text{percep}}=E\left\|\left(\Phi_{\text{VGG}}\left(T_2^{\text{hr}}\right)_{i,j}-\Phi_{\text{VGG}}\left(T_2^f\right)_{i,j}\right)\right\|_1 \tag{7}$$

where β and δ are the coefficients that balance the different loss functions. $\text{Loss}_{\text{percep}}$ calculates the 1-norm distance between T_2^{hr} and T_2^{lr} features obtained by the VGG-19. Perceptual loss uses features before activation, which can provide stronger supervision for brightness consistency and texture recovery

$$\text{Loss}_G^{\text{re}}=-E\left[\log\left(1-D_{\text{re}}\left(T_2^{\text{hr}},T_2^f\right)\right)\right]-E\left[\log\left(D_{\text{re}}\left(T_2^f,T_2^{\text{hr}}\right)\right)\right]. \tag{8}$$

The adversarial loss $\text{Loss}_G^{\text{re}}$ is symmetric and benefits from the gradient of T_2^{hr} and T_2^f in the adversarial training

$$L_{\text{cont}}=E\left\|T_2^f-T_2^{\text{hr}}\right\|_1 \tag{9}$$

where L_{cont} is the content loss to evaluate the 1-normal distance between the SR image T_2^f and the HR image T_2^{hr}.

Compared SR model trained with pixel loss, the SR model trained with perceptual loss, adversarial loss, and content loss training achieves lower PSNR, but they have brought significant gains in perceptual quality [20], [21]. Note that the hyperparameter settings and the effect achieved of the ESR module have been verified in the article [21], [22], and we will only use them here without further corroboration.

Algorithm 1 further describes the flowchart of ESR-DMNet.

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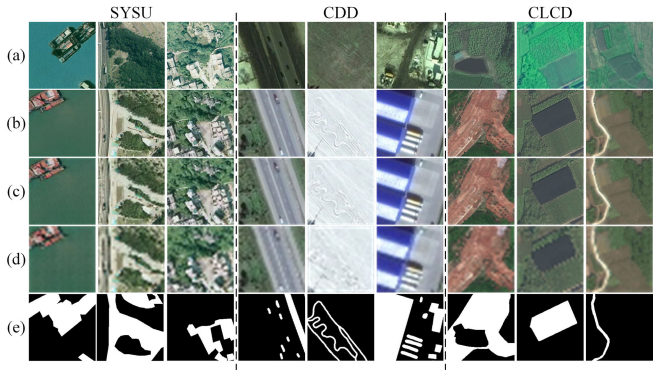


Fig. 5. Examples of different-resolution samples in SYSU, CDD, and CLCD. (a) HR image at time T_1. (b) HR image at time T_2 (1×). (c) 4× LR image at time T_2. (d) 8× LR image at time T_2. (e) Label.

IV. Dataset and Experimental Settings

A. Datasets

Three CD datasets, namely, Sun Yat-Sen University (SYSU) [13], season-varying CD dataset (CDD) [41], and cropland CD dataset (CLCD) [42], are used to evaluate the performance of the proposed method. Table I shows the main properties of the three datasets. We crop the CLCD dataset into image pairs of size 256 × 256.

All the HR images at time T_2 of SYSU, CDD, and CLCD datasets are four and eight times downsampled by BI, respectively, to obtain 4× LR images and 8× LR images at time T_2. This article uses three CD datasets with three resolutions for experimental verification. The examples of different resolution samples in SYSU, CDD, and CLCD are shown in Fig. 5.

B. Comparative Methods

Nine benchmark CD methods based on DL are adopted for comparative analysis. It includes three simple structure CD methods, namely, FC-EF [8], FC-Siam-conc [8], and FC-Siam-diff [8]. Two classification learning RSCD methods, namely, image fusion network (IFN) [9] and SNUNet [10]. Two metric learning RSCD methods, namely, STANet [12] and DASNet. A transformer-based CD method, namely BIT [44]. An SR-based CD method, namely SRCDNet [29].

C. Implementation Details

1) Parameter Setting: All experiments are implemented by Pytorch 1.2.0 and GeForce GTX 1080 Ti. The batch size is set to 16 for the CD of 1× images, and DMNet without ESR is used for CD. For the CD of 4× LR images and 8× LR images at time T_2, batch size is set to 4 and 8, respectively, and ESR-DMNet is used for CD. Adam, as the optimizer, set the learning rate of the DMNet, generator, and relative discriminator as $1e-4$, $1e-3$, and $1e-3$, respectively. Train 100 epochs to converge the model. Residual scaling parameter α is set to 0.2, m to 2, both λ and β to 0.005, and δ to 0.01.

2) CD Evaluation Metrics: Precision (Pre), recall (Rec), and $F1$ are used as evaluation indicators to measure the accuracy of the network. Recall and precision constrain each other, and $F1$ takes both indicators into account

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{10}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{11}$$

$$F1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{12}$$

where TP is the number of changed pixels correctly detected, TN is the number of unchanged pixels correctly detected, FP represents the number of unchanged pixels incorrectly predicted as changed pixels, and FN represents the number of changed pixels incorrectly predicted as unchanged pixels.

3) SR Evaluation Metrics: Image quality evaluation methods include subjective methods based on human perception and objective calculation methods. PSNR is a common objective calculation method used to evaluate and compare SR algorithms. For two images X and Y with size $H \times W$, where X is the reference image and Y is the reconstructed SR image, their MSE is defined as

$$\text{MSE} = \frac{1}{HW}\sum_{i=0}^{H-1}\sum_{j=0}^{W-1}(X(i,j)-Y(i,j))^2. \tag{13}$$

PSNR is defined by the maximum pixel value between images and the MSE

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{\max(Y)^2}{\text{MSE}}\right) \tag{14}$$

where $\max(Y)$ is the maximum value of image Y. The larger PSNR is, the closer Y is to X, and the higher the image quality is.

Since PSNR only cares about the difference between corresponding pixels rather than visual perception, it indicates that PSNR is fundamentally inconsistent with the subjective assessments of human observers. To do this, we introduce a new perceptual index, which is judged by the Ma score [44] and the nonreference indicators of NIQE [45]. The lower the perception index, the better the perceived quality

$$\text{perceptual index} = \frac{1}{2}((10-\text{Ma}) + \text{NIQE}). \tag{15}$$

V. Experimental Results

All comparison methods except SRCDNet and ESR-DMNet require the bitemporal images with the same resolution, so the 4× and 8× LR images at time T_2 will be upsampled to the original image size using BI. Tables II–IV show the accuracy comparisons of three datasets under three resolutions, visualized in Fig. 6. Fig. 7 illustrates the rise and fall of 4× relative to 1× and 8× relative to 4× accuracy of three datasets.

A. Comparison of CD Methods at the Same Resolution

As shown in Table II, compared with the other nine methods, our method has achieved the highest accuracy in three 1× datasets, with $F1$ of 78.40%, 95.93%, and 67.02%, respectively.

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TABLE I
DATASETS' ATTRIBUTES

Datasets	Pairs（Training/Validation/Test）	Size	Source	Resolution (m/Pixel)
SYSU	12000/4000/4000	256 × 256	Aerial images	0.5
CDD	10000/3000/3000	256 × 256	Google Earth images	0.03–1
CLCD	360/120/120	512 × 512	Gaofen-2	0.5–2

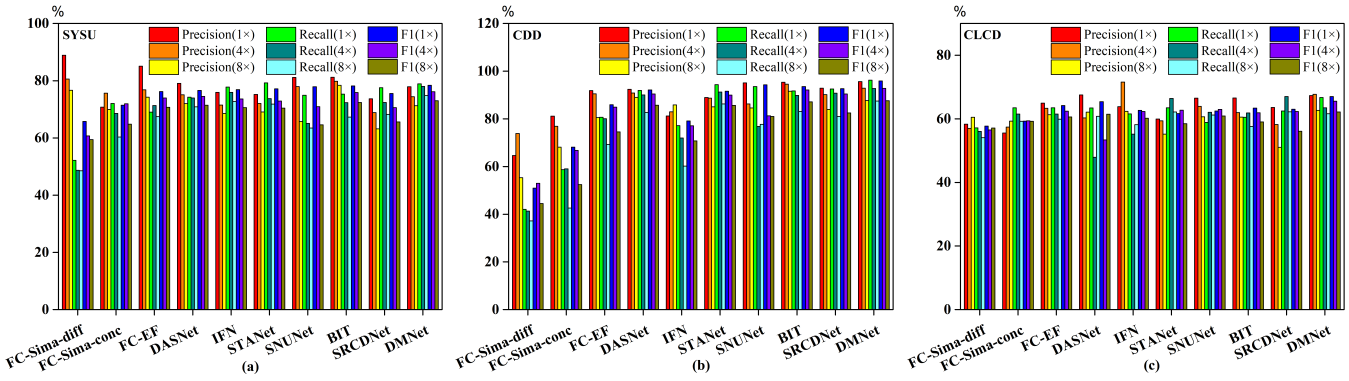


Fig. 6. Comparative visualization of the accuracy of ten methods on (a) SYSU, (b) CDD, and (c) CLCD.

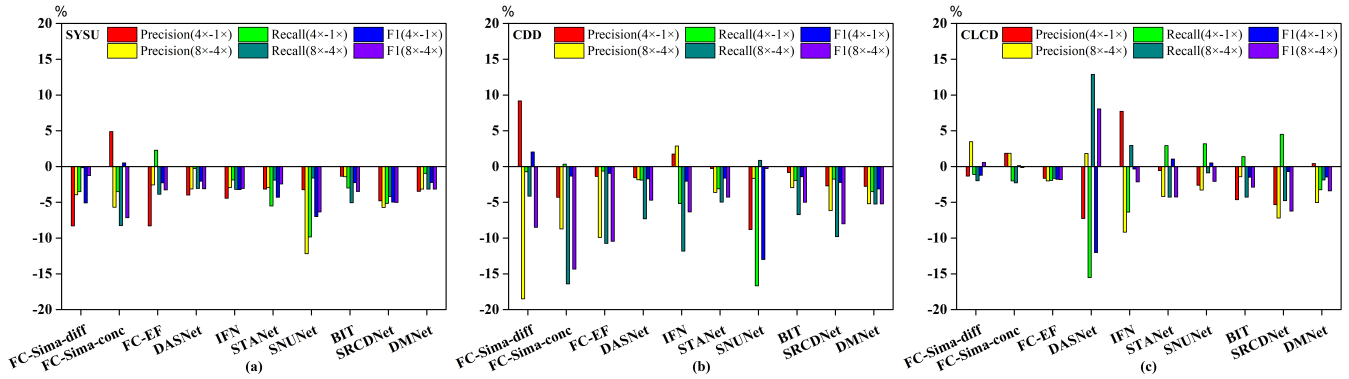


Fig. 7. Rise and fall of 4× relative to 1× and 8× relative to 4× accuracy. (a) SYSU. (b) CDD. (c) CLCD.

TABLE II
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THREE 1× DATASETS

Method	1× SYSU			1× CDD			1× CLCD		
	Pre(%)	Rec(%)	F1(%)	Pre(%)	Rec(%)	F1(%)	Pre(%)	Rec(%)	F1(%)
FC-Sima-diff	**88.90**	52.20	65.78	64.65	42.07	50.97	58.34	57.16	57.74
FC-Sima-conc	70.78	72.08	71.43	81.17	58.69	68.12	55.57	63.50	59.27
FC-EF	85.10	69.02	76.22	91.85	80.68	85.90	64.96	63.46	64.20
DASNet	79.09	74.22	76.58	92.40	91.90	92.15	**67.54**	63.43	65.42
IFN	75.94	77.81	76.86	81.23	77.17	79.15	63.83	61.57	62.68
STANet	75.18	**79.24**	77.16	88.96	94.37	91.58	59.95	63.48	61.66
SNUNet	81.16	74.91	77.91	95.06	93.51	94.27	66.54	58.90	62.49
BIT	81.19	75.34	78.15	95.39	91.74	93.53	66.59	60.51	63.40
SRCDNet	73.66	77.57	75.56	92.84	92.54	92.69	63.57	62.49	63.03
DMNet	77.88	78.92	**78.40**	**95.62**	**96.24**	**95.93**	67.31	**66.73**	**67.02**

1) Comparisons on 1× SYSU: For 1× SYSU, DMNets $F1$ is slightly higher than the second-ranked transformer-based BIT and the third-ranked classification-based SNUNet by 0.25% and 0.49%, respectively, and the same metrics-based STANet achieves the fourth-ranked accuracy and achieves the highest recall, but with an $F1$ that is 1.24% lower than DMNet. Although FC-Sima-diff achieves the highest precision, $F1$ is the lowest, 12.62 % lower than DMNet.

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TABLE III
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THREE 4× DATASETS

Method	4× SYSU			4× CDD			4×CLCD		
	Pre(%)	Rec(%)	F1(%)	Pre(%)	Rec(%)	F1(%)	Pre(%)	Rec(%)	F1(%)
BI+FC-Sima-diff	**80.61**	48.68	60.70	73.84	41.34	53.00	57.00	56.06	56.53
BI+FC-Sima-conc	75.66	68.56	71.94	76.87	59.03	66.78	57.42	61.50	59.39
BI+FC-EF	76.81	71.30	73.95	90.48	80.05	84.95	63.31	61.53	62.41
BI+DASNet	75.11	73.95	74.53	90.85	90.01	90.43	60.29	47.91	53.39
BI+IFN	71.51	75.93	73.65	82.97	71.99	77.09	**71.55**	55.21	62.33
BI+STANet	72.03	73.73	72.87	88.69	91.26	89.95	59.39	66.42	62.71
BI+SNUNet	77.93	65.07	70.92	86.25	76.83	81.27	63.92	62.08	62.99
BI+BIT	79.83	72.34	75.90	**94.55**	89.81	92.12	61.96	61.89	61.92
SRCDNet	68.88	72.41	70.60	90.14	90.78	90.46	58.25	**67.00**	62.32
ESR-DMNet	74.42	**77.98**	**76.16**	92.86	**92.74**	**92.80**	67.74	63.49	**65.55**

TABLE IV
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THREE 8× DATASETS

Method	8× SYSU			8× CDD			8× CLCD		
	Pre(%)	Rec(%)	F1(%)	Pre(%)	Rec(%)	F1(%)	Pre(%)	Rec(%)	F1(%)
BI+FC-Sima-diff	76.68	48.51	59.43	55.34	37.21	44.50	60.48	54.08	57.10
BI+FC-Sima-conc	69.98	60.31	64.79	68.14	42.63	52.44	59.27	59.23	59.25
BI+FC-EF	74.25	67.43	70.68	80.56	69.30	74.51	61.30	59.89	60.59
BI+DASNet	71.98	70.89	71.43	89.00	82.71	85.74	62.13	60.79	61.45
BI+IFN	68.56	72.73	70.58	85.83	60.18	70.75	62.37	58.16	60.19
BI+STANet	69.08	71.84	70.43	85.06	86.28	85.67	55.21	62.15	58.47
BI+SNUNet	65.74	63.46	64.58	84.61	77.69	81.00	60.63	61.19	60.91
BI+BIT	**78.40**	67.27	72.41	**91.60**	83.08	87.13	60.53	57.62	59.04
SRCDNet	63.16	68.22	65.59	83.99	80.98	82.46	51.05	**62.24**	56.09
ESR-DMNet	71.27	**74.83**	**73.01**	87.67	**87.51**	**87.59**	**62.70**	61.62	**62.16**

2) Comparisons on 1× CDD: For 1× CDD, DMNet achieves the highest accuracy in precision, recall, and $F1$, with 95.62%, 96.24%, and 95.93%, respectively. The classification-based SNUNet ranks second, with $F1$ of 94.27%, 1.66% lower than DMNet. The transformer-based BIT ranks third with an $F1$ of 93.53%, which is 2.40% lower than DMNet. SRCDNet achieves fourth place, with $F1$ of 92.69%, 3.24% lower than DMNet.

3) Comparisons on 1× CLCD: For 1× CLCD, the optimal recall and $F1$ are obtained by DMNet, which are 66.73% and 67.02%, respectively. DASNet, also a metric-based RSCD method, achieves the highest precision, 67.54%, $F1$ is 65.42%, ranking second in accuracy, 1.60% lower than DMNet. FC-EF has the third-ranked accuracy, with $F1$ of 64.20%, 2.82% lower than DMNet. It is followed by the BIT and SRCDNet.

B. Comparison of CD Methods at Different Resolutions

1) Comparison of CD Methods at 4× Resolution: As shown in Table III, ESR-DMNet outperforms nine baseline methods on three 4× datasets, with the highest $F1$, which is 76.16%, 92.80%, and 65.55%, respectively.

a) Comparisons on 4× SYSU: Compared with the 1× SYSU, the accuracy of all methods on 4× SYSU decreases to different degrees, except for the FC-Sima-conc. SNUNet decreases the most, with $F1$ dropping by 6.99%. It is followed by FC-Sima-diff with 5.08% decrease in $F1$. ESR-DMNet achieves the optimal accuracy on recall and $F1$ with 77.98% and 76.16%, respectively, and only 2.24% decrease in $F1$ compared to 1× SYSU. Transformer-based BIT ranks second in accuracy, with an $F1$ of 75.90%, slightly lower than ESR-DMNet. DASNet ranks third in accuracy, with an $F1$ of 74.53%, which is 1.63% lower than ESR-DMNet. FC-Sima-diff still achieves the highest precision and lowest $F1$.

b) Comparisons on 4× CDD: Compared with 1× CDD, $F1$ decreases on 4× CDD for all methods except the FC-Sima-diff. SNUNet shows the largest decrease in accuracy, with a 13.00% decrease in $F1$. ESR-DMNet has slightly more $F1$ degradation than the other methods except SNUNet, with 3.13% $F1$ degradation, but ESR-DMNet also achieves the highest accuracy on both recall and $F1$, 92.74% and 92.80%, respectively. Transformer-based BIT ranks second with 92.12% $F1$, which is 0.68% lower than ESR-DMNet. SRCDNet achieves the third ranked accuracy with $F1$ of 90.46%, which is 2.34% lower than ESR-DMNet. This further proves the effectiveness of the SR-based CD network for RS images with different resolutions. The accuracy of the two metric-based methods, DASNet and STANet, follows closely, ranking fourth and fifth, respectively, with $F1$ of 90.43% and 89.95%.

c) Comparisons on 4× CLCD: Compared with the 1× CLCD, the accuracy of most of the methods on 4× CLCD is decreased to varying degrees. DASNet has the highest

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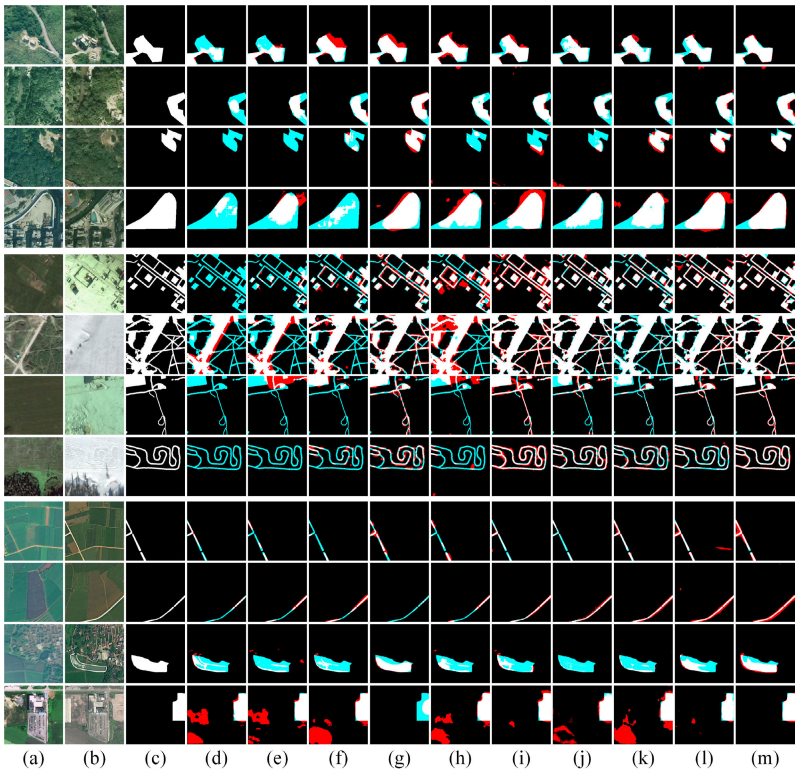


Fig. 8. Visualization of comparative experiments on 1× images (SYSU: rows 1–4; CDD: rows 5–8; and CLCD: rows 9–12). (a) HR image at time T_1. (b) HR image at time T_2. (c) Label. (d) FC-Siam-diff. (e) FC-Siam-conc. (f) FC-EF. (g) DASNet. (h) IFN. (i) STANet. (j) SNUNet. (k) BIT. (l) SRCDNet. (m) DMNet.

accuracy reduction, with $F1$ decreasing by 12.03%. The $F1$ of ESR-DMNet decreases by 1.47%, which is in the middle to upper level, but still achieves the highest accuracy with an $F1$ of 65.55%. SNUNet achieves the second-ranked accuracy with an $F1$ of 62.99%. STANet achieves the third highest accuracy with an $F1$ of 62.71%. Compared with 1× CLCD, the accuracy of SNUNet and STANet is slightly upward, and $F1$ is improved by 0.50% and 1.05%. IFN achieves the highest precision at 71.55%, ranking the fifth, while $F1$ achieves 62.33%. SRCDNet achieves the highest recall at 67.00%, ranking sixth in accuracy and $F1$ at 62.32%. Compared with 1× CLCD, both methods, IFN and SRCDNet, show a decrease in accuracy, with $F1$ decreasing by 0.35% and 0.71%.

2) Comparison of CD Methods at 8× Resolution: As shown in Table IV, ESR-DMNet performs best in three 8× datasets, with $F1$ reaching 73.01%, 87.59%, and 62.16%, respectively.

a) Comparisons on 8× SYSU: Compared with 4× SYSU, the accuracy of all methods on 8× SYSU is reduced, with FC-Sima-conc accuracy decreasing the highest and $F1$ decreasing by 7.15%. SNUNet has the second highest drop in accuracy, with $F1$ dropping 6.34%. The accuracy of SRCD-Net follows with a decrease of 5.01% in $F1$. The $F1$ of ESR-DMNet decreases by 3.15%, with a moderate decrease, and the optimal accuracy is obtained in recall and $F1$, which are 74.83% and 73.01%, respectively. The transformer-based BIT method ranks second in accuracy with an $F1$ of 72.41% and achieves the highest precision of 78.40%. DASNet has the third-ranked accuracy with 71.43% $F1$. Compared with the same SR-based metric CD method SRCDNet and transformer-based BIT, our method performs well on 4× SYSU and 8× SYSU.

b) Comparisons on 8× CDD: Compared with 4× CDD, all methods show a larger $F1$ decrease on 8× CDD except SNUNet, which show a smaller decrease. $F1$ of ESR-DMNet decreases by 5.21%, which is less than the average level, while $F1$ of SRCDNet decreases by 8.00%, which is higher than the average level. FC-Sima-conc shows the largest $F1$ reduction of 14.34%, follows by FC-EF with an $F1$ reduction of 10.44%. Compared with other methods, ESR-DMNet achieves the highest accuracy in recall and $F1$, which are 87.51% and 87.59%, respectively. The second ranked transformer-based BIT achieves the highest precision of 91.60%, but $F1$ is 0.46% lower than ESR-DMNet. The third ranked DASNet has an $F1$ of 85.74% but has an $F1$ of 1.85% lower than ESR-DMNet. STANet achieves the fourth-ranked accuracy, with $F1$ of 85.67%, 1.92% lower than ESR-DMNet. SRCDNet achieves the fifth-ranked accuracy, with $F1$ at 82.46%, 5.13% lower than ESR-DMNet. It also further proves the effectiveness of the metric-based CD method.

c) Comparisons on 8× CLCD: Compared with 4× CLCD, the accuracy of other methods on 8× CLCD is further reduced except for FC-Sima-diff and DASNet. ESR-DMNet achieves the highest accuracy on precision and $F1$, which are 62.70% and 62.16%, respectively, and $F1$ decreases by 3.39% compared with 4× CLCD. DASNet achieves the second-ranked accuracy with 61.45% in $F1$. In particular, compared with 4× CLCD, $F1$ improves by 8.06%. SNUNet ranks third in accuracy with 60.91% in $F1$, and $F1$ is 2.08%

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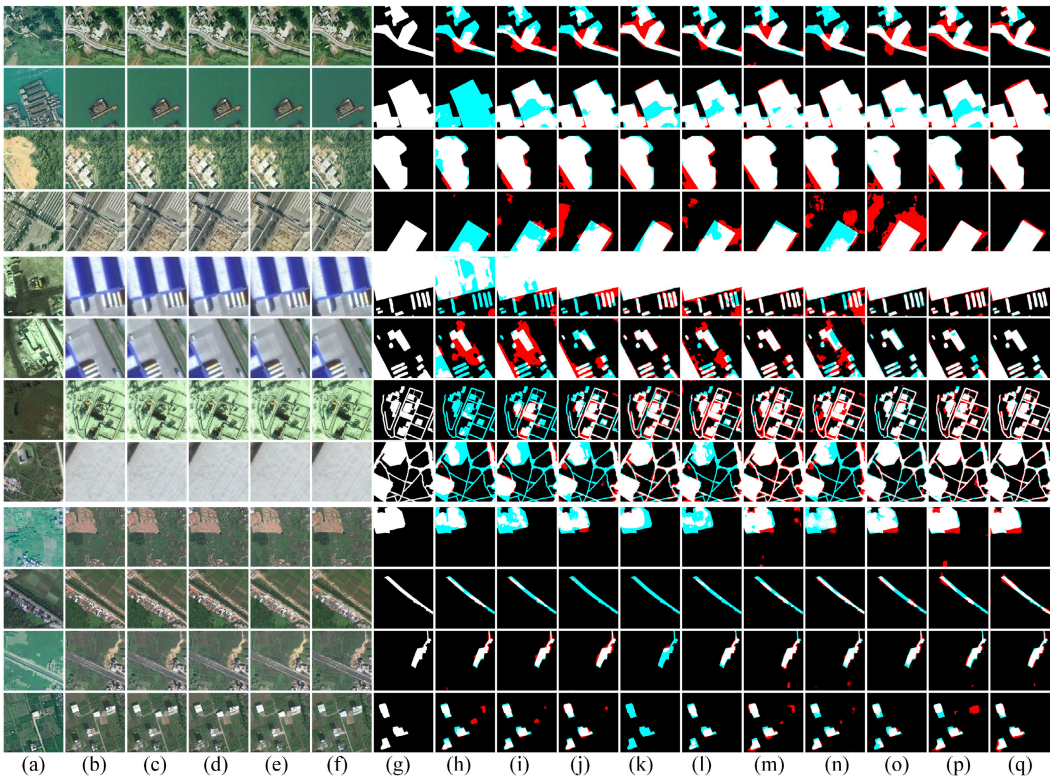


Fig. 9. Visualization of comparative experiments on 4× images (SYSU: rows 1–4; CDD: rows 5–8; and CLCD: rows 9–12). (a) HR image at time T_1. (b) HR image at time T_2. (c) 4× LR image at time T_2. (d) 4× Bicubic image at time T_2. (e) 4× SR image at time T_2 by SRCDNet. (f) 4× ESR image at time T_2 by ESR-DMNet. (g) Label. (h) FC-Siam-diff. (i) FC-Siam-conc. (j) FC-EF. (k) DASNet. (l) IFN. (m) STANet. (n) SNUNet. (o) BIT. (p) SRCDNet. (q) ESR-DMNet.

lower than 4× CLCD. The accuracy of SRCDNet decreases the most, and $F1$ decreases by 6.23%.

C. Visualization

The visualization results of 1×, 4×, and 8× images on SYSU, CDD, and CLCD datasets are shown in Figs. 8–10, respectively, where TP is shown in white, TN in black, FP in red, and FN in blue.

Whether compared with the traditional BI or SRCDNet, our method can obtain sharper edges and more realistic and natural textures on the 4× and 8× SR image of three datasets, but the quality of the 8× SR image is lower than that of the 4× SR image.

1) Visualization of SYSU: Our method performs well with 1×, 4×, and 8× images. The 1× images show suburban dilation, change of vegetation, and newly built urban buildings. The 4× images show road expansion, sea construction, suburban dilation, and groundwork before construction. The 8× images show the change of vegetation, road expansion, suburban dilation, and sea construction.

2) Visualization of CDD: ESR-DMNet can detect smaller changes compared to other methods. The 1× images show the changes in buildings, roads, and land. The 4× images show the changes in cars, buildings, and roads. The 8× images show the changes in cars, roads, and buildings.

3) Visualization of CLCD: The 1× images show the changes in roads, lakes, and bare land. The 4× images show the changes in bare land and buildings. The 8× images show the changes in vegetation and roads. Compared with other methods, ESR-DMNet can detect finer changes compared to other methods.

D. Ablation Study

For the first two ablation experiments, the same SR method plays the same role in influencing the accuracy of different CD methods. Therefore, we use the same CD method and different SR methods in combination with each other for experimental validation. For the third ablation experiment, we conduct ablation experiment on the core modules of DMNet, without involving the ESR module.

1) Ablation Experiment on the Influence of Different SR Methods on the Accuracy of CD Results: Ablation experiments on the influence of different SR methods on the accuracy of CD results are performed on 4× and 8× images of SYSU. Ablation experiments are conducted for the three SR methods, namely, BI, SRM [29] of SRCDNet, and ESR. DMNet is chosen as the CD method. Quantitative results are shown in Table V. ESR-DMNet achieves the highest accuracy on both 4× and 8× image, $F1$ being 76.16% and 73.01%, respectively, 1.65% and 3.40% higher than BI + DMNet and 1.23% and 2.63% higher than SRM + DMNet. The reason for this is partially explained in Fig. 11. With 4× images, our method reconstructs the best, with detailed details and boundary information well recovered. This is followed by SRM, with edge blurring only for fine features. BI has the worst results, with blurring at some of the edges. At 8× image,

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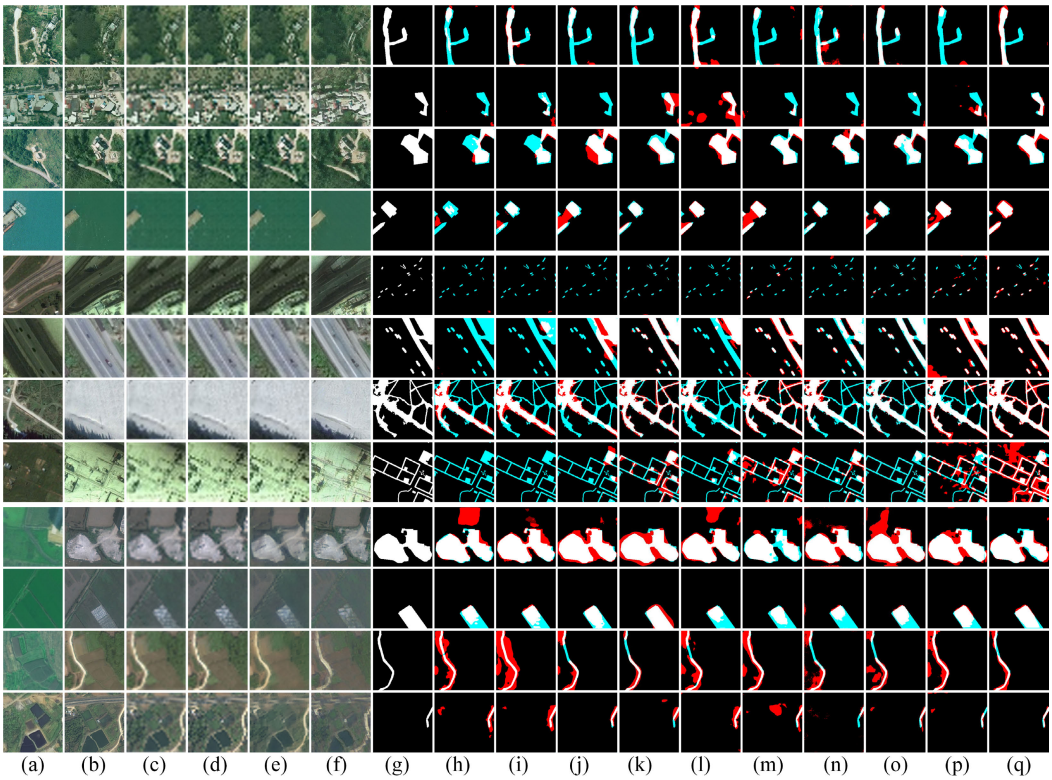


Fig. 10. Visualization of comparative experiments on 8× images (SYSU: rows 1–4; CDD: rows 5–8; and CLCD: rows 9–12). (a) HR image at time T_1. (b) HR image at time T_2. (c) 8× LR image at time T_2. (d) 8× Bicubic image at time T_2. (e) 8× SR image at time T_2 by SRCDNet. (f) 8× ESR image at time T_2 by ESR-DMNet. (g) Label. (h) FC-Siam-diff. (i) FC-Siam-conc. (j) FC-EF. (k) DASNet. (l) IFN. (m) STANet. (n) SNUNet. (o) BIT. (p) SRCDNet. (q) ESR-DMNet.

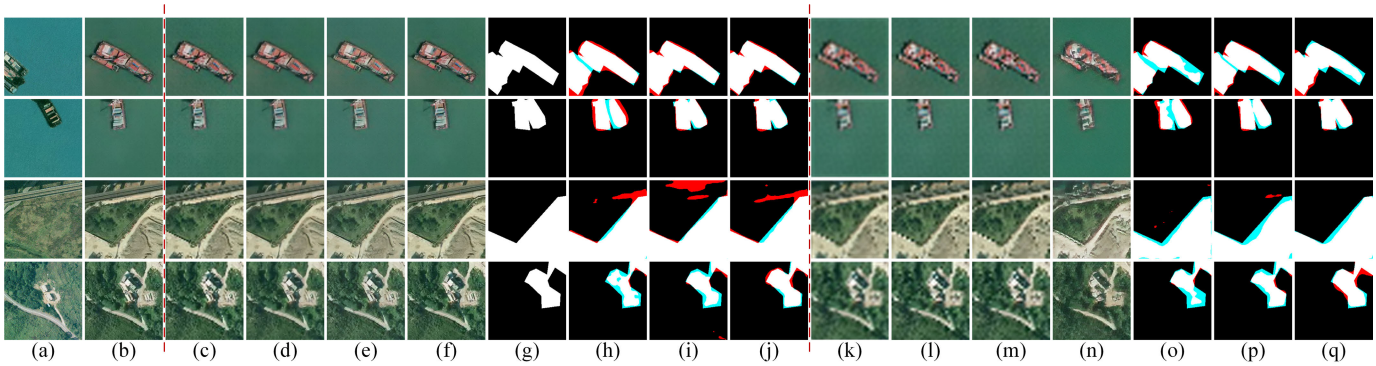


Fig. 11. Ablation experiment on the influence of different SR methods on the accuracy of CD results. (a) HR image at time T_1. (b) HR image at time T_2. (c) 4× LR image at time T_2. (d) 4× Bicubic image at time T_2 by BI. (e) 4× SR image at time T_2 by SRM. (f) 4× ESR image at time T_2 by ESR. (g) Label. (h) 4× BI + DMNet. (i) 4× SRM + DMNet. (j) 4× ESR-DMNet. (k) 8× LR image at time T_2. (l) 8× Bicubic image at time T_2 by BI. (m) 8× SR image at time T_2 by SRM. (n) 8× ESR image at time T_2 by ESR. (o) 8× BI + DMNet. (p) 8× SRM + DMNet. (q) 8× ESR-DMNet.

there is no obvious difference between before and after BI reconstruction, with jagged artefacts and edge blurring, too smooth. SRM is slightly better but also has jagged artefacts and edge blurring. Although our method is less effective than the reconstruction under 4× image, it can still recover part of the boundary information. It can be concluded that the quality of reconstructed images with different SR methods is positively correlated with the accuracy of CD.

2) Comparison of the Quality of Reconstructed Image by Different SR Methods: The images reconstructed by the three SR methods are compared and analyzed in terms of both objective calculation methods and subjective methods based on human perception. The objective calculation methods mainly include the traditional indicator PSNR and the perceptual index, which is consistent with the subjective assessment of human observers. Comparison experiments are performed on 4× and 8× images of SYSU. From the quantitative results in Table VI, it can be seen that for 4× images, the PSNR of BI and SRM is 0.89% and 1.51% higher than that of ESR, respectively, but the ESR has 2.15% and 2.37% lower perception index than BI and SRM. For 8× images, BI and SRM have 1.16% and 1.17% higher PSNR than ESR, but ESR has 4.35% and 4.24% lower perception indicator, respectively. Although the PSNR of our method is slightly lower than that

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TABLE V
ABLATION EXPERIMENTS OF THE INFLUENCE OF DIFFERENT SR METHODS ON THE ACCURACY OF CD RESULTS ON SYSU

Module		4× SYSU			8× SYSU		
SR method	CD method	Pre (%)	Rec (%)	F1 (%)	Pre (%)	Rec (%)	F1 (%)
BI	DMNet	70.92	78.48	74.51	70.19	69.04	69.61
SRM		**75.61**	74.26	74.93	71.13	69.65	70.38
ESR		74.42	**77.98**	**76.16**	**71.27**	**74.83**	**73.01**

TABLE VI
COMPARATIVE EXPERIMENT OF IMAGE QUALITY RECONSTRUCTED BY DIFFERENT SR METHODS ON 4× AND 8× IMAGES OF SYSU

Method	4× SYSU		8× SYSU	
	PSNR	Perceptual index	PSNR	Perceptual index
BI	23.46	7.02	19.66	9.83
SRM	**24.08**	7.24	**19.67**	9.72
ESR	22.57	**4.87**	18.50	**5.48**

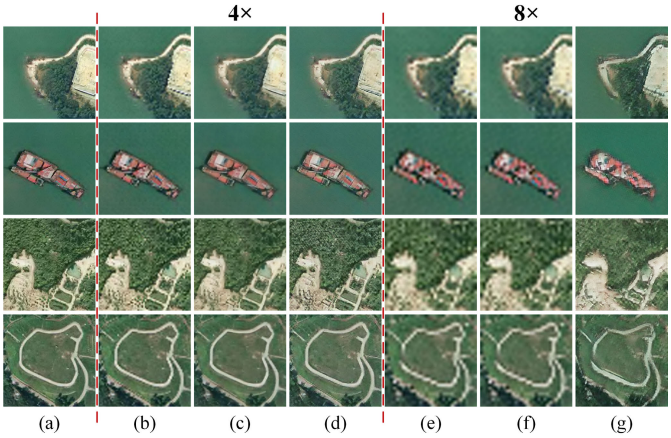


Fig. 12. Comparison of images reconstructed by different SR methods on SYSU. (a) HR image at time T_2. Reconstruction on 4× image at time T_2: (b) BI, (c) SRCDNet, and (d) ESR-DMNet. Reconstruction on 8× image at time T_2: (e) BI, (f) SRCDNet, and (g) ESR-DMNet.

of BI and SRM in the traditional methods, the perceptual index is much lower than that of BI and SRM, indicating that our method improves the reconstruction quality in perception.

PSNR-oriented methods tend to output excessively smooth results without sufficient high-frequency detail. ESR can perceptually improve the reconstruction quality. The visualization results are shown in Fig. 12, and it can be seen that the image reconstructed by BI shows jagged artifacts and blurred edges and is too smooth. The image reconstructed by SRM does not perform well under 8× image and also shows jagged artifacts and blurred edges. The RRDB in the generator in ESR can remove the artifacts effectively, and the relative discriminator in ESR can help to obtain sharper edges and more realistic, natural textures. Although our method is less accurate at 8× images than 4× images, it can achieve sharper edges and more realistic, natural textures.

TABLE VII
CORE MODULE ABLATION EXPERIMENTS ON 1× IMAGES OF SYSU

Module			Pre (%)	Rec (%)	F1 (%)
Spatial detail branch	LRNet	Dual-path FFM			
√	×	√	75.63	77.94	76.77
×	√	×	77.25	77.31	77.28
√	√	×	77.41	78.39	77.90
√	√	√	**77.88**	**78.92**	**78.40**

TABLE VIII
PERFORMANCE AND SPEED COMPARISON

Method	Params. (M)	FLOPs (G)	Time /Epoch(s)	F1 (%)
FC-Sima-diff	1.35	4.73	346	65.78
FC-Sima-conc	1.55	5.33	348	71.43
FC-EF	1.10	2.02	338	76.22
DASNet	48.22	100.72	2330	76.58
IFN	50.44	82.26	1245	76.86
STANet	16.89	6.43	171	77.16
SNUNet	12.03	54.83	971	77.91
BIT	3.50	10.63	400	78.15
SRCDNet(1×)	16.98	35.84	2606	75.56
DMNet	27.99	9.72	568	78.40

3) Ablation Experiment on the Core Modules of DMNet: We perform pruning tests on the core modules of DMNet, including the LRNet in the semantic branch, the spatial detail branch, and the dual-path FFM. DMNet without LRNet, DMNet without spatial detail branch and dual-path FFM, and DMNet without dual-path FFM are designed. As can be seen from Table VII, compared with DMNet, the accuracy of the three ablation experiments decreases, and $F1$ decreases by 1.63%, 1.12%, and 0.50%, respectively. Specifically, the precision and recall of DMNet without LRNet have different degrees of decreases, 2.25%, 0.98%, respectively, and more decreases in precision. This is because the function of LRNet is to obtain more HR semantic information. The precision and recall of DMNet without spatial detail branch and dual-path FFM decrease by 0.63% and 1.61%, respectively, and the recall has decreased more. The spatial detail branch can encode rich spatial information. The precision and recall of DMNet without dual-path FFM decrease by nearly the same amount,

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TABLE IX
EFFICIENCY OF TWO DL SR-BASED CD NETWORK FOR RS IMAGES WITH DIFFERENT RESOLUTIONS

Method	4× Params. (M)	4× FLOPs (G)	4× Time /Epoch (s)	4× F1 (%)	8×Params. (M)	8× FLOPs (G)	8× Time /Epoch(s)	8× F1 (%)
SRCDNet	**22.93**	**47.80**	4092	70.60	**23.08**	46.65	4776	65.59
ESR-DMNet	44.87	68.71	**2756**	**76.16**	44.87	**33.41**	**2030**	**73.01**

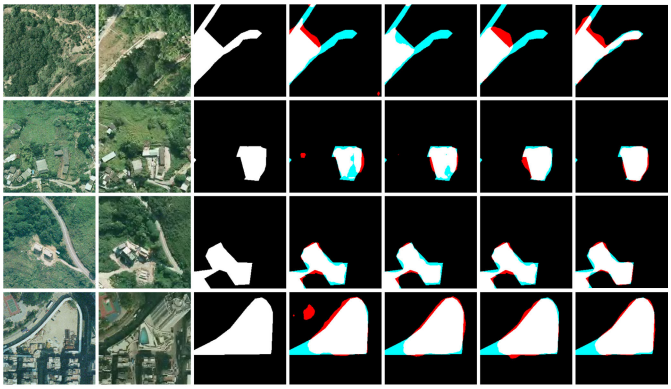


Fig. 13. Core module ablation experiments on 1× images of SYSU. (a) HR image at time T_1. (b) HR image at time T_2. (c) Label. (d) DMNet without LRNet. (e) DMNet without spatial detail branch and dual-path FFM. (f) DMNet without dual-path FFM. (g) DMNet.

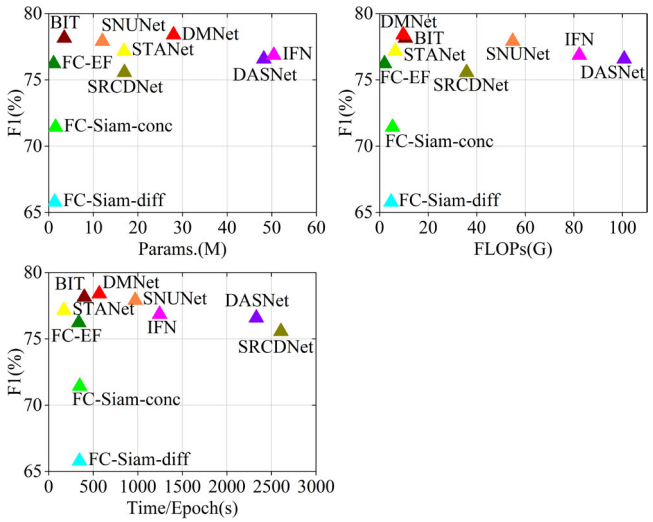


Fig. 14. Model efficiency comparison. (a) Params. (b) FLOPs. (c) Training time of an epoch.

by 0.47% and 0.53%, respectively. The function of dual-path FFM is that it can select and combine the features output from the spatial detail branch and the features output from the semantic branch. The visualization of the results is shown in Fig. 13.

E. Model Efficiency

Table VIII shows the model efficiency of the ten methods, which are calculated under 1× images, and Fig. 14 visualizes the performance. The image input size for computing model parameters (Params.) and floating-point operations per second (FLOPs) is $1 \times 3 \times 256 \times 256$. The image input size for computing the training time of an epoch in 1× image of SYSU is $8 \times 3 \times 256 \times 256$. It can be seen that the three FCN-based models, FC-Siam-diff, FC-Siam-conc, and FC-EF, have the lowest FLOPs and parameters, but the CD accuracies are all lower than our model. Our model has fewer model parameters than DASNet and IFN. FLOP is at the medium level of 9.72 G, which is better than DASNet, IFN, SNUNet, BIT, and SRCDNet. Although the parameters and FLOPs of DMNet are higher than those of STANet, the accuracy of DMNet is improved by 1.24%. The training time of DMNet is almost at the medium level. BI method is simple, but the reconstructed images tend to generate overly smooth images with blurred edges and jagged artefacts, which affects the accuracy of CD. We add the comparison of the efficiency of two DL SR-based CD network for RS images with different resolutions, SRCDNet and ESR-DMNet, under 4× and 8× images on SYSU, as shown in Table IX. Our method has higher parameters and FLOPs than SRCDNet under 4× images and the higher parameters and lower FLOPs than SRCDNet under 8× images. However, the training time is much lower than SRCDNet under 4× and 8× images, and the $F1$ is 5.56% and 7.42% higher than SRCDNet, respectively. Therefore, relatively speaking, our method can achieve high-accuracy and high-efficiency CD.

VI. CONCLUSION

In this article, we propose an ESR-DMNet for RS images with different resolutions. A novel ESR module is provided for the CD of RS images with different resolutions, which helps SR images learn sharper edges and more detailed textures and solves the spatial resolution mismatch problem in CD of different resolution RS images. An effective and efficient DMNet is proposed, which can process shallow spatial details separately from deep semantic information, so as to achieve high-accuracy and high-efficiency CD. Three loss functions are used to optimize the DMNet, the relative discriminator and the generator in the ESR module, respectively. Compared with nine state-of-the-art DL methods, our method shows good performance at three resolutions on SYSU, CDD, and CLCD datasets. Ablation comparison experiments are designed from three aspects, namely, the effect of different SR methods on the accuracy of CD results, the quality of reconstructed images from different SR methods, and the core modules of DMNet, as well as comparing the model efficiency, to further demonstrate the effectiveness of the proposed method. Due to the fact that this article requires truth HR images corresponding to LR images for supervised learning, its practicability is further limited. In the future, we will explore unsupervised DL-based CD of RS images with different resolutions.

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